

- 6-15. In the network of the figure, the switch K is closed and a steady state is reached in the network. At $t = 0$, the switch is opened. Find an expression for the current in the inductor, $i_2(t)$.

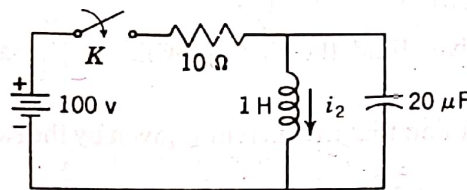


Fig. P6-15.

- 6-16. The capacitor of the figure has an initial voltage $v_C(0-) = V_1$, and at the same time the current in the inductor is zero. At $t = 0$, the switch K is closed. Determine an expression for the voltage $v_2(t)$.

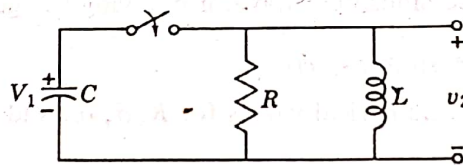


Fig. P6-16.

- 6-17. The voltage source in the network of the figure is described by the equation, $v_1 = 2 \cos 2t$ for $t \geq 0$ and is a short circuit prior to that time. Determine $v_2(t)$. Repeat if $v_1 = K_1 t$ for $t \geq 0$ and $v_1 = 0$ for $t < 0$.

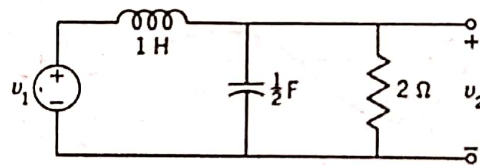


Fig. P6-17.

- 6-18. Solve the following nonhomogeneous differential equations for $t \geq 0$.

(a) $\frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + i = 1$

(b) $\frac{d^2 i}{dt^2} + 3 \frac{di}{dt} + 2i = 5t$

(c) $\frac{d^2 i}{dt^2} + 3 \frac{di}{dt} + 2i = 10 \sin 10t$

(d) $\frac{d^2 q}{dt^2} + 5 \frac{dq}{dt} + 6q = te^{-t}$

(e) $\frac{d^2 v}{dt^2} + 5 \frac{dv}{dt} + 6v = e^{-2t} + 5e^{-3t}$

- 6-19. Solve the differential equations given in Prob. 6-18 subject to the following initial conditions:

$$x(0+) = 1 \quad \text{and} \quad \frac{dx}{dt}(0+) = -1$$

where x is the general dependent variable.

- 6-20. Find the particular solutions to the differential equations of Prob. 6-18 for the following initial conditions:

$$x(0+) = 2 \quad \text{and} \quad \frac{dx}{dt}(0+) = -1$$

where x is the dependent variable in each case.

- 6-21. Solve the differential equation

$$2 \frac{d^3 i}{dt^3} + 9 \frac{d^2 i}{dt^2} + 13 \frac{di}{dt} + 6i = K_0 t e^{-t} \sin t$$

which is valid for $t \geq 0$, if $i(0+) = 1$, $di/dt(0+) = -1$, and $d^2 i/dt^2(0+) = 0$.

- 6-22. A special generator has a voltage variation given by the equation $v(t) = t$ V, where t is the time in seconds and $t \geq 0$. This generator is connected to an RL series circuit, where $R = 2 \Omega$ and $L = 1$ H, at time $t = 0$ by the closing of a switch. Find the equation for the current as a function of time $i(t)$.

- 6-23. A bolt of lightning having a waveform which is approximated as $v(t) = t e^{-t}$ strikes a transmission line having resistance $R = 0.1 \Omega$ and inductance $L = 0.1$ H (the line-to-line capacitance is assumed negligible). An equivalent network is shown in the accompanying diagram. What is the form of the current as a function of time? (This current will be in amperes per unit volt of the lightning; likewise the time base is normalized.)

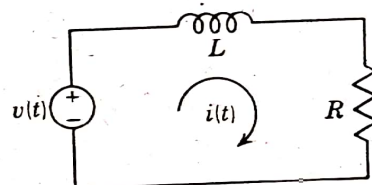


Fig. P6-23.

- 6-24. In the network of the figure, the switch K is closed at $t = 0$ with the capacitor initially unenergized. For the numerical values given, find $i(t)$.

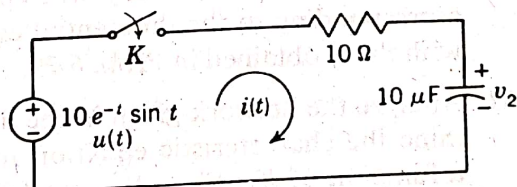


Fig. P6-24.

- 6-25. In the network shown in the accompanying figure, a steady state is reached with the switch K open. At $t = 0$, the switch is closed. For the element values given, determine the current, $i(t)$ for $t \geq 0$.

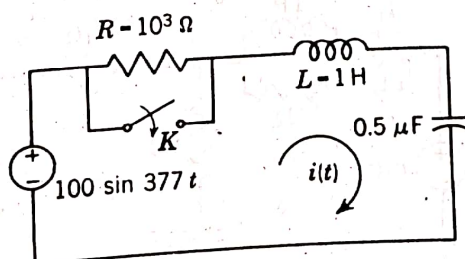


Fig. P6-25.

- 6-26. In the network shown in Fig. P6-25, a steady state is reached with the switch K open. At $t = 0$, the value of the x resistor R is changed to the critical value, R_{cr} defined by Eq. (6-88). For the element values given, determine the current $i(t)$ for $t \geq 0$.

- 6-27. Consider the network shown in Fig. P6-24. The capacitor has an initial voltage, $v_C = 10$ V. At $t = 0$, the switch K is closed. Determine $i(t)$ for $t \geq 0$.
- 6-28. The network of the figure is operating in the steady state with the switch K open. At $t = 0$, the switch is closed. Find an expression for the voltage, $v(t)$ for $t \geq 0$.

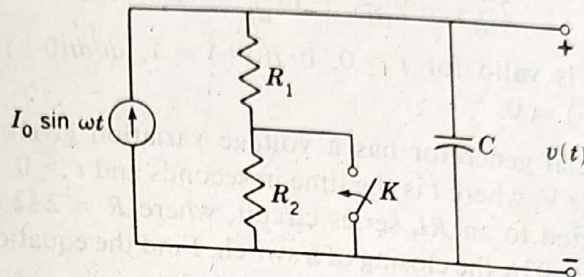


Fig. P6-28.

- 6-29. Consider a series RLC network which is excited by a voltage source. (a) Determine the characteristic equation corresponding to the differential equation for $i(t)$. (b) Suppose that L and C are fixed in value but that R varies from 0 to ∞ . What will be the locus of the roots of the characteristic equation? (c) Plot the roots of the characteristic equation in the s plane if $L = 1$ H, $C = 1$ μ F, and R has the following values: 500 Ω , 1000 Ω , 3000 Ω , 5000 Ω .
- 6-30. Consider the RLC network of Prob. 6-16. Repeat Prob. 6-29, except that in this case the study will concern the characteristic equation corresponding to the differential equation for $v_2(t)$. Compare results with those obtained in Prob. 6-29.
- 6-31. Analyze the network given in the figure on the loop basis, and determine the characteristic equation for the currents in the network as a function of K_1 . Find the value(s) of K_1 for which the roots of the characteristic equation are on the imaginary axis of the s plane. Find the range of values of K_1 for which the roots of the characteristic equation have positive real parts.

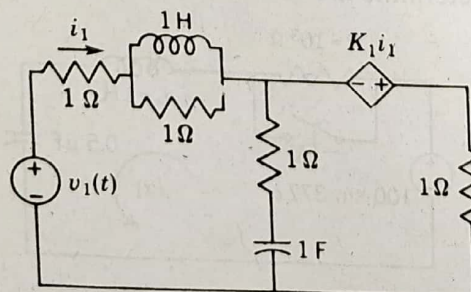


Fig. P6-31.

- 6-32. Show that Eq. (6-121) can be written in the form

$$i = Ke^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t + \phi)$$

Give the values for K and ϕ in terms of K_5 and K_6 of Eq. (6-121).

- 6-33. A switch is closed at $t = 0$ connecting a battery of voltage V with a series RL circuit. (a) Show that the energy in the resistor as a function of time is

$$w_R = \frac{V^2}{R} \left(t + \frac{2L}{R} e^{-Rt/L} - \frac{L}{2R} e^{-2Rt/L} - \frac{3L}{2R} \right) \text{ joules}$$

(b) Find an expression for the energy in the magnetic field as a function of time. (c) Sketch w_R and w_L as a function of time. Show the steady-state asymptotes, that is, the values that w_R and w_L approach as $t \rightarrow \infty$. (d) Find the total energy supplied by the voltage source in the steady state.

- 6-34. In the series RLC circuit shown in the accompanying diagram, the frequency of the driving force voltage is

- (1) $\omega = \omega_n$ (the undamped natural frequency)
- (2) $\omega = \omega_n \sqrt{1 - \zeta^2}$ (the natural frequency)

These frequencies are applied in two separate experiments. In each experiment we measure (a) the peak value of the transient current when the switch is closed at $t = 0$, and (b) the maximum value of the steady-state current. (a) In which case (that is, which frequency) is the maximum value of the transient greater? (b) In which case (that is, which frequency) is the maximum value of the steady-state current greater?

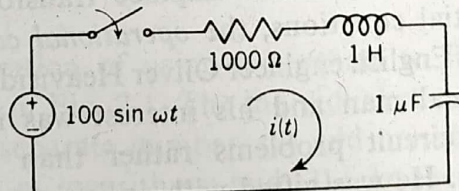


Fig. P6-34.

$$R_2 - R_2 e^{-R_1 t/L}. \quad 4-13. (b) i = (Vt/L)e^{-R_1 t/L}. \quad 4-16. (h)$$

$$i = \frac{[(a-1) \sin 2t - 2 \cos t]e^{-t} + 2e^{-at}}{(a-1)^2 + 4}$$

$$4-17. i_L = 0.6 + 0.067e^{-3.57t}. \quad 4-20. (a) R = 10^4 \Omega, (b) C = 2.5 \mu F.$$

$$4-21. (b) i = 0.01(1 - e^{-16t}) \text{ amp.} \quad 4-22. w_C/w_T = (\frac{1}{2})(1 - e^{-t/RC}).$$

Chapter 5

5-1. 0.1, -100, 100,000. 5-3. $d^2i/dt^2(0+) = -9 \times 10^5 \text{ amp/sec}^2$ 5-5. $-8 \text{ V/sec}^2, 26 \text{ V/sec}^3$. 5-6. 0 V/sec². 5-8. 100, $-10^4, 10^6$. 5-13. (b) 0, $R_2 V/(R_1 + R_2)$, (c) V/CR_1 , 0, (d) $R_2 V/R_1 LC$. 5-15. $d^2v_1/dt^2(0+) = V_0/R$. 5-18. $1.406 \times 10^{-5} \text{ amp/sec}^2$ 5-20. (b) 66.7 V, (c) 3.33, 1.67 amps, (d) 33.3, -83,300, 5-21. $R_1 i(0+)$, 0, $R_1[(di/dt)(0+) - Ri(0+)/L]$, 0. 5-25. (a) VR_1/R_2 , (b) $(1/C) - R_1$. 5-27. (a) 0, $v(0+)/R_1 C_1$, (b) 0, 0, (c) 0, 0. 5-28. (a) $v_K(0+) = VR_3/(R_1 + R_2 + R_3)$.

Chapter 6

6-3. (f) $i = (A \cos \sqrt{1.75}t + B \sin \sqrt{1.75}t)e^{-t/2}$. (h) $i = Ae^{-2t} + Bte^{-2t}$. 6-5. (a) $i = 2e^{-t} - e^{-2t}$. (b) $i = 3e^{-2t} - 2e^{-3t}$. (f) $i = e^{-0.5t}(\cos 1.32t + 0.379 \sin 1.32t)$. (h) $i = e^{-2t} + te^{-2t}$. 6-9. $i = Ae^{-4.18t} + e^{-1.12t}(B \cos 1.06t + C \sin 1.06t)$. 6-11. $t_1 = 1/\alpha$. 6-14. $i = e^{-140t} \sin 1570t$. 6-15. $i_L = 10 \cos 316t$. 6-18. (b) $i = K_1 e^{-t} + K_2 e^{-2t} - \frac{15}{4} + (5t/2)$, (e) $v = (K_1 + t)e^{-2t} + (K_2 - 5t)e^{-3t}$. 6-23. $i = 5t^2 e^{-t}$. 6-24. $i = 10^{-4} e^{-t}(\cos t - \sin t) - 10^{-4} e^{-10t}$. 6-28. $i = 10^{-3}(20.3 \cos 377t + 14.8 \sin 1414t - 0.8 \cos 1414t) \text{ amp.}$ 6-29. $s(500) = -250 \pm j968$, $s(1000) = -500 \pm j866$, $s(3000) = -2,620, -380$, $s(5000) = -4,790, -210$. 6-31. Characteristic equation: $(5 - K_1)s^2 + (6 - 2K_1)s + (2 - K_1) = 0$; the system is stable for $K_1 < 2$. 6-33. (d) $w_S = w_R + w_L = (V^2/R)[t + (L/R)(e^{-Rt/L} - 1)]$.

Chapter 7

7-9. $F_1(s) = 2/[s(s^2 + 4)]$. 7-16. $F_8(s) = s^2/[(s^2 + a^2)^2]$. 7-22. $(V/R_1)e^{-(R_1+R_2)t/L}$. 7-24. $(V/R) \cos(t/\sqrt{LC})$. 7-30. (a) $K_0 = \frac{3}{8}$, $K_2 = \frac{1}{4}$, $K_4 = \frac{3}{8}$. 7-35. $i = K_1 e^t + K_2 e^{-t} - 25 + (\frac{1}{3})e^{2t}$. 7-37. $i = (\frac{1}{3})t^3 + 2e^{-t} + 2$. 7-42. $v_a = 13(e^{-0.0215t} - e^{-0.1285t})$. 7-43. 16.7 sec. 7-44. 14 V in 29.2 sec. 7-45. (a) $3.33 + 1.21e^{-6.35t} - 4.5e^{-23.6t}$. 7-46. $1020(e^{-0.155t} - e^{-0.645t})$. 7-49. $v_a = 4 + e^{-0.75t}(-1.5 \cos 0.25t + 0.5 \sin 0.25t)$. 7-50. $i_2 = -5 - 1.34e^{-0.707t} + 16.3e^{0.707t}$.

Chapter 8

8-1. $u(t-2) - u(t-3) + u(t-7) - u(t-8)$. 8-9. $v = K_0[t^2 u(t) - 4(t-1)u(t-1) - (t-2)^2 u(t-2)]$. 8-10. Partial answer: $v = be^{(\ln(b/a)t/c} \cdot u(t) - \dots$. 8-14. (c) $i = [5 \times 10^5 t - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t) - [5 \times 10^5(t-10^{-6}) - 1.25 + 1.25e^{-4 \times 10^5(t-10^{-6})}]u(t-10^{-6})$. 8-17. $i(t) = [1 - e^{-(t-1)}]u(t-1) + [1 - e^{-(t-2)}]u(t-2) + [1 - e^{-(t-3)}]u(t-3) + [1 - e^{-(t-4)}]u(t-4) - 4[1 - e^{-(t-5)}]u(t-5)$. 8-19. $F_1(s) = K[(1/s^2) + (t_0/s)]$. 8-26. $V(s) = (1/s)[e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} - 4e^{-5s}]$. 8-31. $F(s) = 1/[s(1 - e^{-s})]$. 8-33. $F(s) = \{1/[(s + \frac{1}{2})(1 + e^{-(0.5+s)})]\}$. 8-34. (a) $f(t) = te^{at}$.